Method of Multipliers

- Equal-ratio property: if $\dfrac{A_1}{B_1} = \dfrac{A_2}{B_2} = \dfrac{A_3}{B_3}$, then each equals $\dfrac{\ell A_1 + mA_2 + nA_3}{\ell B_1 + mB_2 + nB_3}$.
- Hence for any multipliers ℓ, m, n (constants or functions of x, y, u):
$$\frac{dx}{a} = \frac{dy}{b} = \frac{du}{c} = \frac{\ell\, dx + m\, dy + n\, du}{\ell a + mb + nc}.$$
- If possible choose ℓ, m, n with $\ell a + mb + nc = 0$.
- Then the numerator also vanishes: $\ell\, dx + m\, dy + n\, du = 0$.
- If this is an exact differential $d\varphi$, integrating gives $\varphi(x, y, u) = c_1$.
- The numbers / functions ℓ, m, n are called Lagrange's multipliers.



Method of Multipliers (contd.)

- Repeat with another set ℓ', m', n' to get $\psi(x, y, u) = c_2$.
- Variant: choose ℓ, m, n so that the numerator $\ell\, dx + m\, dy + n\, du$ is the exact differential of the denominator $\ell a + mb + nc$.
- Then equating with another ratio gives a logarithmic integral.
- Common choices: $(1, 1, 1)$, $(1, -1, 0)$, (x, y, u), $\left(\frac{1}{x}, \frac{1}{y}, \frac{1}{u}\right)$.
- Look at the structure of a, b, c (symmetry, cyclic patterns) to guess multipliers.
- General solution: $G(\varphi, \psi) = 0$, or $\varphi = g(\psi)$, or $\psi = h(\varphi)$.
- Grouping and multipliers may be mixed in the same problem.

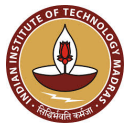


Why Multipliers Work

- Suppose $\varphi(x, y, u) = c_1$ is a first integral; then $d\varphi = 0$:

$$\varphi_x\, dx + \varphi_y\, dy + \varphi_u\, du = 0. \tag{1}$$

- Along characteristics, $dx = a\,\mu$, $dy = b\,\mu$, $du = c\,\mu$ for some $\mu(x, y, u)$.
- Substituting in (1): $a\,\varphi_x + b\,\varphi_y + c\,\varphi_u = 0$.
- So the multipliers $(\ell, m, n) = (\varphi_x, \varphi_y, \varphi_u)$ satisfy $\ell a + mb + nc = 0$.
- and $\ell\, dx + m\, dy + n\, du = d\varphi$ is exact.
- Conversely, such multipliers with an exact numerator produce a first integral.
- Finding multipliers $\Leftrightarrow$ finding the gradient of a first integral.

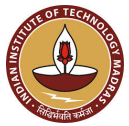


Example: $(\beta u - \gamma y)\, p + (\gamma x - \alpha u)\, q = \alpha y - \beta x$

- (α, β, γ are given constants).
- Auxiliary equations: $\dfrac{dx}{\beta u - \gamma y} = \dfrac{dy}{\gamma x - \alpha u} = \dfrac{du}{\alpha y - \beta x}$.
- Multipliers x, y, u:

$$x(\beta u - \gamma y) + y(\gamma x - \alpha u) + u(\alpha y - \beta x) = 0.$$

- Hence $x\,dx + y\,dy + u\,du = 0$.
- $\Rightarrow d\left(\dfrac{x^2 + y^2 + u^2}{2}\right) = 0 \Rightarrow x^2 + y^2 + u^2 = c_1$.



Example (contd.)

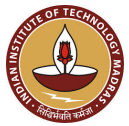
- Multipliers α, β, γ:

$$\alpha(\beta u - \gamma y) + \beta(\gamma x - \alpha u) + \gamma(\alpha y - \beta x) = 0.$$

- Hence $\alpha\, dx + \beta\, dy + \gamma\, du = 0$.
- $\Rightarrow \alpha x + \beta y + \gamma u = c_2$.
- General solution:

$$G\left(x^2 + y^2 + u^2,\ \alpha x + \beta y + \gamma u\right) = 0.$$

- Geometry: characteristics are circles (sphere $\cap$ plane).
- Integral surfaces are surfaces of revolution about the line through 0 along (α, β, γ).

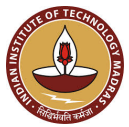


Example: $(xu - y)\,p + (yu - x)\,q = 1 - u^2$

- Auxiliary equations: $\dfrac{dx}{xu - y} = \dfrac{dy}{yu - x} = \dfrac{du}{1 - u^2}$.
- Multipliers $(1, 1, 0)$ for the first fraction:

$$\frac{dx + dy}{(xu - y) + (yu - x)} = \frac{du}{1 - u^2}.$$

- $\Rightarrow \dfrac{d(x + y)}{(x + y)(u - 1)} = \dfrac{du}{(1 - u)(1 + u)}$.
- $\Rightarrow \dfrac{d(x + y)}{x + y} = -\dfrac{du}{1 + u}$.
- Integrating: $\ln(x + y) = -\ln(1 + u) + \ln c_1$.
- $\Rightarrow (x + y)(1 + u) = c_1$.



Example (contd.)

- Multipliers $(1, -1, 0)$:

$$\frac{dx - dy}{(xu - y) - (yu - x)} = \frac{du}{1 - u^2}.$$

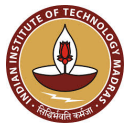
- $\Rightarrow \dfrac{d(x-y)}{(x-y)(u+1)} = \dfrac{du}{(1-u)(1+u)}.$
- $\Rightarrow \dfrac{d(x-y)}{x-y} = \dfrac{du}{1-u}.$
- Integrating:
  $\ln(x-y) = -\ln(1-u) + \ln c_2 \Rightarrow (x-y)(1-u) = c_2.$
- General solution:

$$G\big((x+y)(1+u),\ (x-y)(1-u)\big) = 0.$$



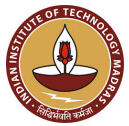
Example: $u(x+y)\,p + u(x-y)\,q = x^2 + y^2$

- Auxiliary equations: $\dfrac{dx}{u(x+y)} = \dfrac{dy}{u(x-y)} = \dfrac{du}{x^2+y^2}$.
- Multipliers $(y, x, -u)$: $uy(x+y) + ux(x-y) - u(x^2+y^2) = 0$.
- $\Rightarrow y\,dx + x\,dy - u\,du = 0 \Rightarrow d\left(xy - \dfrac{u^2}{2}\right) = 0$.
- First integral: $2xy - u^2 = c_1$.
- Multipliers $(x, -y, -u)$:
  $ux(x+y) - uy(x-y) - u(x^2+y^2) = 0$.
- $\Rightarrow x\,dx - y\,dy - u\,du = 0 \Rightarrow x^2 - y^2 - u^2 = c_2$.
- General solution: $G\left(2xy - u^2,\ x^2 - y^2 - u^2\right) = 0$.



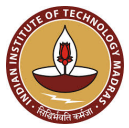
Practice Set I (Multipliers)

- Find the general solutions:
- (1) $(y-u)\,p+(u-x)\,q=x-y$.
- (2) $x(y-u)\,p+y(u-x)\,q=u(x-y)$.
- (3) $\dfrac{(b-c)}{a}\,yu\,p+\dfrac{(c-a)}{b}\,ux\,q=\dfrac{(a-b)}{c}\,xy$.
- (4) $x(y^2-u^2)\,p+y(u^2-x^2)\,q=u(x^2-y^2)$.
- (5) $[y(x+y)+\kappa u]\,p+[x(x+y)-\kappa u]\,q=u(x+y)$ (κ constant).
- (6) $x(y^2-u^2)\,p-y(u^2+x^2)\,q=u(x^2+y^2)$.
- Hint: try $(1,1,1)$, (x,y,u), $(\frac{1}{x},\frac{1}{y},\frac{1}{u})$.



Practice Set II (Grouping / Mixed)

- Find the general solutions:
- (1) $(x+u)\,p + y\,q = u + y^2$.
- (2) $\left(\dfrac{u^2 - 2yu - y^2}{x}\right)p + (y+u)\,q = y - u$.
- (3) $2xu\,p + 2yu\,q = u^2 - x^2 - y^2$.
- (4) $(y^2+u^2)\,p - xy\,q = -xu$.
- (5) $\cos(x+y)\,p + \sin(x+y)\,q = u$.
- (6) $(x^2+2y^2)\,p - xy\,q = xu$.
- (7) $(1+y)\,p + (1+x)\,q = u$.



Equivalent Ratio (Eigenvalue) Method

- Useful when a, b, c are linear in x, y, u.
- Each ratio equals $\dfrac{\ell\, dx + m\, dy + n\, du}{\ell a + mb + nc}$.
- Seek ℓ, m, n and λ with

$$\ell a + mb + nc = \lambda\,(\ell x + my + nu).$$

- Then $\dfrac{d(\ell x + my + nu)}{\lambda\,(\ell x + my + nu)}$ equals every ratio.
- Comparing coefficients of x, y, u gives a homogeneous linear system for (ℓ, m, n).
- A non-trivial solution requires a determinant to vanish: a cubic for λ.
- Each root λ_i gives $w_i = \ell_i x + m_i y + n_i u$ with $\dfrac{dw_i}{\lambda_i w_i}$ equal to the common ratio.

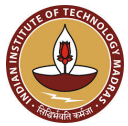


Equivalent Ratio Method: Integrals

- From three roots we get

$$\frac{dw_1}{\lambda_1 w_1} = \frac{dw_2}{\lambda_2 w_2} = \frac{dw_3}{\lambda_3 w_3}.$$

- Equate the first two: $\lambda_2 \dfrac{dw_1}{w_1} = \lambda_1 \dfrac{dw_2}{w_2}$.
- Integrate: $w_1^{1/\lambda_1} w_2^{-1/\lambda_2} = c_1$.
- Similarly from the first and third: $w_1^{1/\lambda_1} w_3^{-1/\lambda_3} = c_2$.
- General solution: $G\left(w_1^{1/\lambda_1} w_2^{-1/\lambda_2},\ w_1^{1/\lambda_1} w_3^{-1/\lambda_3}\right) = 0$.
- Roots may be complex; real integrals can then be formed by combination.
- This is the same as diagonalising the coefficient matrix.



Example:
$(x+y-u)\,p+(y+u-x)\,q=u+x-y$

- Auxiliary equations: $\dfrac{dx}{x+y-u}=\dfrac{dy}{y+u-x}=\dfrac{du}{u+x-y}.$
- Each ratio equals

$$\frac{\ell\,dx+m\,dy+n\,du}{\ell(x+y-u)+m(y+u-x)+n(u+x-y)}. \tag{1}$$

- We want this to be of the form

$$\frac{\ell\,dx+m\,dy+n\,du}{\lambda\,(\ell x+my+nu)}. \tag{2}$$

- Compare coefficients of x, y, u in the denominators of (1) and (2).



Example (contd.): The Characteristic Equation

- Coefficients: $\ell - m + n = \lambda\ell$, $\ell + m - n = \lambda m$, $-\ell + m + n = \lambda n$.
- i.e.

$$\begin{aligned}(1-\lambda)\ell - m + n &= 0\\ \ell + (1-\lambda)m - n &= 0\\ -\ell + m + (1-\lambda)n &= 0.\end{aligned} \tag{3}$$

- Non-trivial solution: $\begin{vmatrix} 1-\lambda & -1 & 1 \\ 1 & 1-\lambda & -1 \\ -1 & 1 & 1-\lambda \end{vmatrix} = 0.$
- $\Rightarrow \lambda^3 - 3\lambda^2 + 6\lambda - 4 = 0 \Rightarrow (\lambda - 1)(\lambda^2 - 2\lambda + 4) = 0.$
- $\lambda = 1$, $\lambda = 1 \pm i\sqrt{3}$, i.e. $\lambda = -2\omega$, $-2\omega^2$ (ω = cube root of unity).

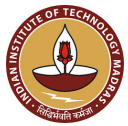


Example (contd.): Multipliers

- $\lambda = 1$: from (3), $m = n$ and $\ell = n$, so $\ell = m = n$; take $(1, 1, 1)$.
- $\lambda = -2\omega$: first two equations of (3) give

$$\ell(1 + 2\omega) - m + n = 0, \qquad \ell + m(1 + 2\omega) - n = 0.$$

- Solving: $\dfrac{\ell}{\omega} = \dfrac{m}{\omega^2} = \dfrac{n}{1}$; take $(\omega, \omega^2, 1)$.
- $\lambda = -2\omega^2$: similarly $\dfrac{\ell}{\omega^2} = \dfrac{m}{\omega} = \dfrac{n}{1}$; take $(\omega^2, \omega, 1)$.
- Use $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$ in the simplifications.
- $w_1 = x + y + u$, $w_2 = \omega x + \omega^2 y + u$, $w_3 = \omega^2 x + \omega y + u$.



Example (contd.): General Solution

- From $\lambda = 1$ and $\lambda = -2\omega$:

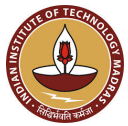
$$\frac{d(x+y+u)}{x+y+u} = \frac{d(\omega x + \omega^2 y + u)}{(-2\omega)(\omega x + \omega^2 y + u)}.$$

- $\Rightarrow (x+y+u)\,(\omega x + \omega^2 y + u)^{1/(2\omega)} = c_1.$
- From $\lambda = 1$ and $\lambda = -2\omega^2$:

$$(x+y+u)\,(\omega^2 x + \omega y + u)^{1/(2\omega^2)} = c_2.$$

- General solution:

$$G\left((x+y+u)(\omega x+\omega^2 y+u)^{\frac{1}{2\omega}},\ (x+y+u)(\omega^2 x+\omega y+u)^{\frac{1}{2\omega^2}}\right) = 0.$$



Remark: A Real First Integral

- Complex integrals can be combined into real ones.
- Here a direct real integral is available using multipliers (x, y, u):
- $x(x+y-u)+y(y+u-x)+u(u+x-y)=x^2+y^2+u^2$.
- So $\dfrac{x\,dx+y\,dy+u\,du}{x^2+y^2+u^2}=\dfrac{d(x+y+u)}{x+y+u}$.
- $\Rightarrow \frac{1}{2}\ln(x^2+y^2+u^2)=\ln(x+y+u)+\text{const.}$
- Real first integral: $\dfrac{x^2+y^2+u^2}{(x+y+u)^2}=c_1$.
- Note $|w_2|^2=x^2+y^2+u^2-xy-yu-ux$, consistent with the above.
- Always check whether simple multipliers give a quicker real answer.



Practice: Equivalent Ratio Method

- Solve by the equivalent ratio method:
- (1) $(y+u)\,p+(u-x)\,q=u+5y$.
- (2) $(x+y-2u)\,p+(y+u-2x)\,q=u+x-2y$.
- (3) $(2x-y)\,p+(2y-u)\,q=2u-x$.
- Steps: write the coefficient matrix, find λ, find (ℓ, m, n) for each λ.
- Form $w_i=\ell_i x+m_i y+n_i u$ and integrate $\dfrac{dw_i}{\lambda_i w_i}=\dfrac{dw_j}{\lambda_j w_j}$.
- Check your multipliers by substituting back into $\ell a+mb+nc=\lambda w$.



Lagrange's Method: n Independent Variables

- Let $u = u(x_1, \ldots, x_n)$ and $u_i = \partial u / \partial x_i$.
- Quasi-linear PDE:

$$a_1 u_1 + a_2 u_2 + \cdots + a_n u_n = c.$$

- $a_i = a_i(x_1, \ldots, x_n, u)$, $c = c(x_1, \ldots, x_n, u)$.
- Characteristic equations:

$$\frac{dx_1}{a_1} = \frac{dx_2}{a_2} = \cdots = \frac{dx_n}{a_n} = \frac{du}{c}.$$

- Find n independent first integrals $\varphi_j(x_1, \ldots, x_n, u) = c_j$, $j = 1, \ldots, n$.
- General solution: $G(\varphi_1, \ldots, \varphi_n) = 0$, G arbitrary of n arguments.
- Or $\varphi_j = g(\varphi_1, \ldots, \varphi_{j-1}, \varphi_{j+1}, \ldots, \varphi_n)$.



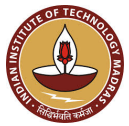
Example: $(y-z)u_x + (z-x)u_y + (x-y)u_z = 0$

- Here $u = u(x, y, z)$.
- Characteristic equations:

$$\frac{dx}{y-z} = \frac{dy}{z-x} = \frac{dz}{x-y} = \frac{du}{0}.$$

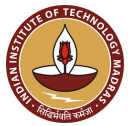
- $du = 0 \Rightarrow u = c_1$.
- Multipliers $(1, 1, 1)$: denominator $= 0 \Rightarrow dx + dy + dz = 0 \Rightarrow x + y + z = c_2$.
- Multipliers (x, y, z): denominator $= 0 \Rightarrow x\,dx + y\,dy + z\,dz = 0 \Rightarrow x^2 + y^2 + z^2 = c_3$.
- General solution:

$$u = f\left(x + y + z,\ x^2 + y^2 + z^2\right), \quad f \text{ arbitrary.}$$



Example: $x\,u_x + y\,u_y + z\,u_z = 2u$

- Characteristic equations: $\dfrac{dx}{x} = \dfrac{dy}{y} = \dfrac{dz}{z} = \dfrac{du}{2u}$.
- $\dfrac{dx}{x} = \dfrac{dy}{y} \Rightarrow \dfrac{y}{x} = c_1$.
- $\dfrac{dx}{x} = \dfrac{dz}{z} \Rightarrow \dfrac{z}{x} = c_2$.
- $\dfrac{dx}{x} = \dfrac{du}{2u} \Rightarrow \dfrac{u}{x^2} = c_3$.
- General solution: $G\left(\frac{y}{x}, \frac{z}{x}, \frac{u}{x^2}\right) = 0$.
- Equivalently $u = x^2\, f\left(\frac{y}{x}, \frac{z}{x}\right)$.
- These are the homogeneous functions of degree 2 in (x, y, z).



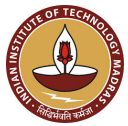
The Cauchy Problem

- Consider $a(x, y, u)\, p + b(x, y, u)\, q = c(x, y, u)$.
- Prescribe u on a curve, e.g. $u(x, y_0) = f(x)$ or $u(x_0, y) = g(y)$.
- More generally u is given on a curve Γ in the xy-plane.
- The data define a space curve C (the initial curve).
- Cauchy problem: find the integral surface that passes through C.
- Such a condition is called an initial condition.
- Geometric idea: through every point of C draw the characteristic.
- The union of these characteristics is the required surface.



Solving the Cauchy Problem via First Integrals

- Step 1: Find first integrals $\varphi(x, y, u) = c_1$, $\psi(x, y, u) = c_2$.
- Step 2: Impose the initial curve: e.g. put $y = y_0$ and $u = f(x)$ into both.
- Step 3: Eliminate x (the curve parameter) to get a relation $H(c_1, c_2) = 0$.
- Step 4: Replace $c_1 \to \varphi(x, y, u)$, $c_2 \to \psi(x, y, u)$.
- Result: $H\big(\varphi(x, y, u), \psi(x, y, u)\big) = 0$ is the required integral surface.
- Step 5: Solve for u if possible, and verify the PDE and the data.
- This picks one particular G from the general solution $G(\varphi, \psi) = 0$.



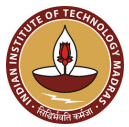
Cauchy Problem 1: $p + q = u$, $u(x, 0) = \frac{x}{2} + e^x$

- Auxiliary equations: $\dfrac{dx}{1} = \dfrac{dy}{1} = \dfrac{du}{u}$.
- $dx = dy \Rightarrow x - y = c_1$ (1).
- $dy = \dfrac{du}{u} \Rightarrow u = c_2 e^y$, i.e. $ue^{-y} = c_2$ (2).
- On $y = 0$: from (1), $x = c_1$; from (2), $u(x, 0) = c_2$.
- Given data: $c_2 = \dfrac{x}{2} + e^x = \dfrac{c_1}{2} + e^{c_1}$.
- Replace c_1, c_2: $ue^{-y} = \dfrac{x - y}{2} + e^{x-y}$.
- Solution: $u(x, y) = \dfrac{(x - y)}{2} e^y + e^x$.



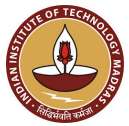
Cauchy Problem 2: $p + y\,q = 2xu$, $u(0, y) = y - y^2$

- Auxiliary equations: $\dfrac{dx}{1} = \dfrac{dy}{y} = \dfrac{du}{2xu}$.
- $dx = \dfrac{dy}{y} \Rightarrow x = \ln y - \ln c_1 \Rightarrow ye^{-x} = c_1$ (1).
- $dx = \dfrac{du}{2xu} \Rightarrow x^2 = \ln u - \ln c_2 \Rightarrow ue^{-x^2} = c_2$ (2).
- On $x = 0$: $y = c_1$ and $u(0, y) = c_2$.
- Data: $c_2 = y - y^2 = c_1 - c_1^2$.
- Hence $ue^{-x^2} = ye^{-x} - y^2e^{-2x}$.
- Solution: $u(x, y) = y\,e^{x^2-x} - y^2\,e^{x^2-2x}$.



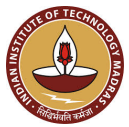
Cauchy Problem 3: $x\,p + y\,q = u,\ \ u(1, y) = y^2$

- Auxiliary equations: $\dfrac{dx}{x} = \dfrac{dy}{y} = \dfrac{du}{u}$.
- $\dfrac{dx}{x} = \dfrac{dy}{y} \Rightarrow \dfrac{y}{x} = c_1$ (1).
- $\dfrac{dx}{x} = \dfrac{du}{u} \Rightarrow \dfrac{u}{x} = c_2$ (2).
- On $x = 1$: $y = c_1$ and $u(1, y) = c_2$.
- Data: $c_2 = y^2 = c_1^2$.
- Hence $\dfrac{u}{x} = \dfrac{y^2}{x^2}$.
- Solution: $u(x, y) = \dfrac{y^2}{x}$ (valid for $x > 0$).
- Check: $x(-y^2/x^2) + y(2y/x) = y^2/x = u$ ✓.



Cauchy Problem 4: $x\,p + q = x(1+u)$, $u(e, y) = y - 2$

- Auxiliary equations: $\dfrac{dx}{x} = \dfrac{dy}{1} = \dfrac{du}{x(1+u)}$.
- $\dfrac{dx}{x} = dy \Rightarrow \ln x - y = c_1$ (1).
- $\dfrac{dx}{x} = \dfrac{du}{x(1+u)} \Rightarrow dx = \dfrac{du}{1+u} \Rightarrow \ln(1+u) - x = c_2$ (2).
- On $x = e$: from (1), $y = 1 - c_1$; from (2), $1 + u = e^{e+c_2}$.
- Data $u = y - 2$: $1 + u = y - 1 = -c_1$.
- Hence $-c_1 = e^{e+c_2}$, i.e. $c_1 = -e^e\, e^{c_2}$.



Cauchy Problem 4 (contd.)

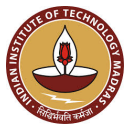
- Replace $c_1 = \ln x - y$ and $e^{c_2} = (1+u)e^{-x}$:

$$\ln x - y = -(1+u)\, e^{e-x}.$$

- Solving for u:

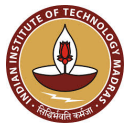
$$u(x,y) = -1 + (y - \ln x)\, e^{x-e}.$$

- Check data: $x = e$ gives $u = -1 + (y-1) = y - 2$ ✓.
- Check PDE: $u_x = \big(-\frac{1}{x} + y - \ln x\big)e^{x-e}$, $u_y = e^{x-e}$.
- $xu_x + u_y = x(y - \ln x)e^{x-e} = x(1+u)$ ✓.



Cauchy Problem 5: $p + \frac{q}{2y} = 2 + u$, $u(x, 0) = \sin x - 2$

- Auxiliary equations: $\dfrac{dx}{1} = \dfrac{dy}{1/(2y)} = \dfrac{du}{2+u}$.
- $dx = 2y\,dy \Rightarrow x - y^2 = c_1$ (1).
- $2y\,dy = \dfrac{du}{2+u} \Rightarrow \ln(2+u) - y^2 = c_2$ (2).
- On $y = 0$: $x = c_1$ and $\ln(2+u) = c_2$.
- Data: $2 + u(c_1, 0) = \sin c_1 = e^{c_2}$.
- Hence $\sin(x - y^2) = (2+u)\,e^{-y^2}$.
- Solution: $u(x, y) = -2 + e^{y^2} \sin(x - y^2)$.



Cauchy Problem 6: $x\,u_y = y\,u_x,\quad u(0,y) = y^2$

- Rewrite: $y\,p - x\,q = 0$.
- Auxiliary equations: $\dfrac{dx}{y} = \dfrac{dy}{-x} = \dfrac{du}{0}$.
- $x\,dx + y\,dy = 0 \Rightarrow x^2 + y^2 = c_1; \qquad du = 0 \Rightarrow u = c_2$.
- General solution: $u = g(x^2 + y^2)$.
- Data: $g(0 + y^2) = y^2 \Rightarrow g(s) = s$.
- Solution: $u(x,y) = x^2 + y^2$.
- The surfaces of revolution about the u-axis again appear.



Cauchy Problem 7: Non-uniqueness

- Find an integral surface of $x\,u_y = y\,u_x$ containing the circle $u = 1,\ x^2 + y^2 = 4$.
- General solution: $G(x^2 + y^2,\ u) = 0$.
- The surface contains the circle iff $G(4, 1) = 0$.
- $G(s, t) = s - 4t$: $x^2 + y^2 = 4u$ (paraboloid).
- $G(s, t) = s - t - 3$: $u = x^2 + y^2 - 3$.
- $G(s, t) = s + t^2 - 5$: $x^2 + y^2 + u^2 = 5$ (sphere).
- Infinitely many integral surfaces contain the given curve.
- Reason: the circle is itself a characteristic curve (see the existence theorem).

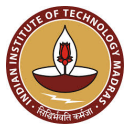


Cauchy Problem 8:
$u(x+y)\,p + u(x-y)\,q = x^2 + y^2$, $u = 0$ on $y = 2x$

- From the earlier example: $2xy - u^2 = c_1$, $x^2 - y^2 - u^2 = c_2$.
- Write the general solution as $x^2 - y^2 - u^2 = f(2xy - u^2)$.
- On the line: $y = 2x$, $u = 0$:

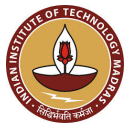
$$x^2 - 4x^2 = f(4x^2) \Rightarrow f(4x^2) = -3x^2.$$

- Hence $f(s) = -\frac{3}{4}s$.
- So $x^2 - y^2 - u^2 = -\frac{3}{4}(2xy - u^2)$.
- $\Rightarrow 4(x^2 - y^2 - u^2) + 3(2xy - u^2) = 0$.
- Solution: $7u^2 = 4(x^2 - y^2) + 6xy$.



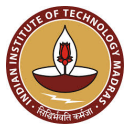
Homework: Cauchy Problems

- Solve the following Cauchy problems:
- (1) $x\,p + q = 1, \quad u(1, y) = e^{-y}$.
- (2) $p + y\,q = u, \quad u(x, 1) = x\,e^{-x}$.
- (3) $x\,p + y^{-1}q = 1, \quad u(x, 0) = 5 - x$.
- (4) $x(y^2 + u)\,p - y(x^2 + u)\,q = (x^2 - y^2)\,u, \quad u(x, -x) = 1$.
- (5) $yu\,p + xu\,q = xy, \quad u(x, 0) = 2x^2$.
- (6) $x^2(y - u)\,p + y^2(u - x)\,q = u^2(x - y), \quad u(x, 1) = -x$.
- (7) $(x - y)\,y^2 p + (y - x)\,x^2 q = (x^2 + y^2)\,u, \quad u(x, 0) = \frac{\kappa^3}{x}$, $\kappa \in \mathbb{R}$.



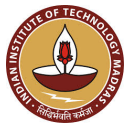
More Practice

- (8) $u_x + u_y = u^2, \quad u(x,0) = \cos x.$
- (9) $y\,u_x + x\,u_y = 0, \quad u(0,y) = e^{-y^2}.$
- (10) $u_x + 2u_y = 0, \quad u(0,y) = 4e^{-2y}.$
- (11) $u_t + u\,u_x = 0, \quad u(x,0) = x$ (show $u = x/(1+t)$).
- (12) Find the surface satisfying $p + q = 1$ through $u = 0$ on $x + y = 1$.



Summary

- First order PDEs: linear $\subset$ semilinear $\subset$ quasi-linear $\subset$ nonlinear.
- Eliminating n constants from $f(x, y, u, a, b) = 0$ gives a first order PDE.
- Eliminating an arbitrary function from $G(\varphi, \psi) = 0$ gives a quasi-linear PDE.
- Solutions: complete, general, singular and particular integrals.
- Integral surfaces are tangent to the vector field (a, b, c).
- Auxiliary equations $\dfrac{dx}{a} = \dfrac{dy}{b} = \dfrac{du}{c}$ define characteristic curves.
- General solution $G(\varphi, \psi) = 0$ via grouping, multipliers or equivalent ratios.
- Cauchy problem: unique solution when the initial curve is non-characteristic.



Looking Ahead: Second Order Linear PDEs

- General second order linear PDE in two variables:

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = H.$$

- $A, \ldots, H$ are functions of x, y, not all zero at any point.
- $u_{tt} = k^2 u_{xx}$ one-dimensional wave equation.
- $u_t = k^2 u_{xx}$ one-dimensional heat equation.
- $u_{xx} + u_{yy} = 0$ two-dimensional Laplace equation.
- $u_{xx} + u_{yy} = f(x, y)$ two-dimensional Poisson equation.
- $u_{tt} = k^2(u_{xx} + u_{yy})$ two-dimensional wave equation.
- Characteristics will again play the central role in their classification.

